

What approaches have been used to address the problem in the project?

- Deep learning and AI to detect URLs
 - https://albahnsen.wordpress.com/wp-content/uploads/2018/05/deepphish-simulating-malicious-ai_submitted.pdf
- More broadly, content analysis (which is computationally complex)
 - Catching Classical and Hijack-Based Phishing Attacks
 - https://link.springer.com/chapter/10.1007/978-3-319-13841-1_18 338
- Phishing URL detection can be done via proactive or reactive means. On the reactive end, we find services such as Google Safe Browsing API2
 - Deepphish article
- These types of services expose a blacklist of malicious URLs to be queried. The blacklists are constructed using different techniques, including manual reporting, honeypots, or by crawling the web in search of known phishing characteristics [5], [6]. For example, browsers make use of blacklists to block access upon reaching the URLs contained within them. One drawback of such a reactive method is that in order for a phishing URL to be blocked, it must be previously included in the blacklist. This implies that web users remain at risk until the URL is submitted and the blacklist is updated. What is more, since the majority of phishing sites are active for less than a day [6], [7], their mission is complete by the time they are discovered and added to the blacklist.
 - Deepphish article
 - With AI you can auto create them super quick now (mimicking sites, tools like Social Engineer Toolkit, etc.)
 - Doesn't work because URLs haven't been up long enough to be identified and will only be up for 1 - 3 days (insert real numbers)
- Networking often works the same way where they don't trust certain IPs or whatever but again you need access to a list
- URL detection is language impartial, and lightweight since you don't have to look at the website
 - <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9759382>
- As pointed out in [28], "phishing patterns evolve constantly, and it is usually hard for a detection method to achieve a high true positive rate while maintaining a low false positive rate." Existing phishing site detection methods fall into one of the following categories: URL matching against human-verified blacklists, heuristics used with machine learning, password based, or some combination of information extraction with information retrieval. The first category of methods has very low false positives rate, but such methods are not robust against future cases. Sheng et al. [21] show that zero-hour protection of major blacklist-based toolbars has true positive rates in the 15-40 % range. Updating these blacklists 320 T. Thakur and R. Verma typically requires heavy human effort and is a slow process. For example, January 2012 statistics from Phishtank show that the median time to verify that a URL is a phish was two hours. Of course, human-verification can be easily overwhelmed by automatically generated URLs. Heuristics used with machine learning suffer from the need for a clean, labeled training corpus, over-fitting to the training corpus and the need for retraining because of model drift over time. Password-based schemes lack robustness for phishing detection [9].

Information extraction based schemes such as named-entity extraction [28] suffer from the lack of parsable sentences on web sites and also the limits of automatic natural language processing techniques. Moreover, named-entity extraction techniques are also particularly prone to lower-casing problems such as failure to recognize paypal (note the lower case) as a named entity.

- file:///Users/a250-808-9416/Downloads/978-3-319-13841-1.pdf
- Different research groups have studied this problem from various perspectives: server-side and browser-side strategies, education/training, and evaluation of anti-phishing tools, detection schemes and finally studies that analyze the reasons behind the success of phishing attacks, e.g., [15,21,7,6,5,3,4,8,1,30].
- https://link.springer.com/chapter/10.1007/978-3-319-13841-1_18 338

4.1 U1: Targets in URL

Cyber criminals put the target (e.g., PayPal, eBay, BankOfAmerica, etc.) in the URL to disguise the URL to the user. According to quarterly reports by APWG (Anti Phishing Working Group), 45-50% phishing URLs contained target names in the URL last year (2013). This strategy is used to fool naive users into believing that they are really visiting the desired page. Many researchers discussed this strategy and use of this feature. We have improved this method over existing techniques. We take the URL, and remove the main domain and TLD from it. We are now left with either subdomain and/or the extended URL. We check targets in such remaining part of the URL and not the complete URL directly. We need an accurate and comprehensive list of targets for this purpose. We analyzed 12,000 URLs from Phishtank and collected the top targets that were targeted more than 0.1% of the time among those 12,000 URLs. As expected, PayPal ranked first with more than 13.6% followed by AOL, which was just 1.1%. Also, phishers try to attack a big population and hence a popular domain. So, we added all the whitelisted domains to the list of targets. This made the list strong and removing the actual domain name with TLD from the input URL helped us to get rid of false positives. Experiments were done to test this function and the result varied from true positive rate of 40% to 50% with almost 0% false positives.

4.2 U2: Misplaced TLDs

This is similar to checking targets in URL, but we check for list of all top level domains (TLDs) that are placed not at the actual TLD location. Phishers use this strategy to disguise URLs. We remove the main domain and actual TLD from the URL and check for the TLDs in the rest of the URL.

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- file:///Users/a250-808-9416/Downloads/978-3-319-13841-1.pdf

- [illegible]

How are these approaches different?

What are their strengths and limitations?

- Url detection can be language agnostic
 - https://link.springer.com/chapter/10.1007/978-3-319-13841-1_18 338
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Where does your project fit among these existing approaches?

General Notes

- The trends in patterns of the phishing sites keep changing and hence, we need a robust algorithm that is not just specific to the given dataset. Also, the lifetime of phishing sites is very short. "On an average a phishing domain lasts 3 days 31 minutes and 8 seconds" [17]
 - https://www.usenix.org/legacy/event/leet08/tech/full_papers/mcgrath/mcgrath.pdf

History of scamming

- <https://ieeexplore.ieee.org/abstract/document/10543498>
- <https://www.taylorfrancis.com/chapters/edit/10.4324/9781843929680-11/scams-simon-mackenzie>
- Telecommunications scams history and lessons learned
 - https://www.tandfonline.com/doi/pdf/10.1198/TECH.2009.08136?casa_token=QfpSkiLNiAgAAAAA:rV9yprNCJZ8jfXlrJ6iaBp5FC-Xy4u-B424NtLjiiR60VQN7se1-r8l8w7lnZJnAOyO4hMXjzk
 - Comes from latin Fraudem, meaning deceit or injury
 - In early days of communication, would convince the telephone operator to give you a free call
- Social engineer toolkit: ability to purchase scamming kits online, whole market and industry there

History of Phishing

- <https://en.wikipedia.org/wiki/Phishing>

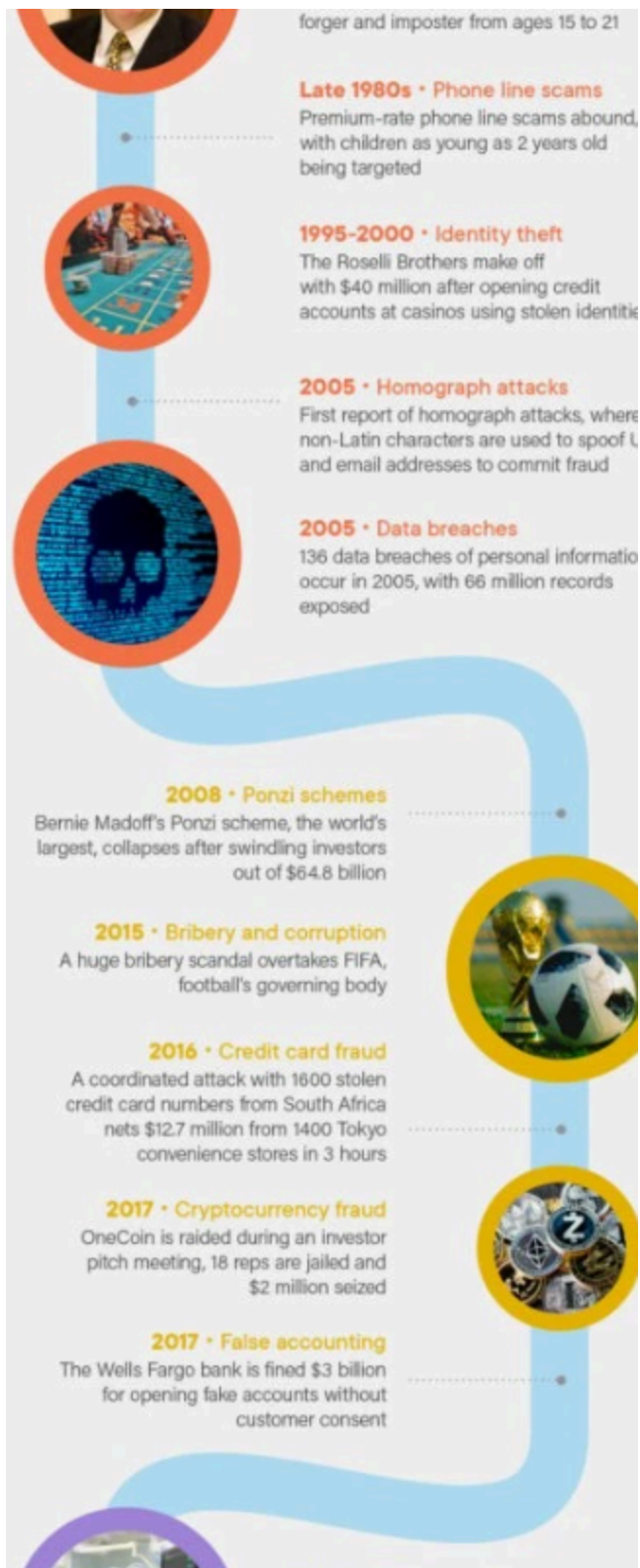
Fraud

- First documented case of fraud was 3400 years ago in 300 BC when two great shipping merchants attempted to conduct insurance fraud. Hegestratos
- Ancient egyptian government struggled to collect taxes in 525 bce
- The expansion of the internet has created more opportunities for fraud. Cybercrime has grown to humongous proportions. No business or individual is exempt from the threat of cyber fraud. As more and more innovations continue to be developed using technology, the rate of fraud increases and gets more sophisticated.
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- The use of technology to perpetrate fraud dates back to the 1980s when the internet started to gain relevance. One of the first scams exploited loopholes in

telecommunications policies by trying to con people into making calls to expensive phone numbers. Then fraudsters began to target people using TV ads, in one instance, little kids were targeted using dial-tone signals on TV.

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- By the early 90s, fraud took on new dimensions when eCommerce platforms became a thing. Fraudsters took advantage of the weaknesses of online credit card use. At the time, credit card verification technology was still in its infancy, so they managed to steal the credit card information of celebrities to purchase expensive items. This was only the beginning of a tactic that would continue to wreak havoc even today.

<https://www.getcybersafe.gc.ca/en/resources/history-phishing> history of phishing



<https://www.trulioo.com/blog/fraud-prevention/history-fraud>

for opening fake accounts without
customer consent

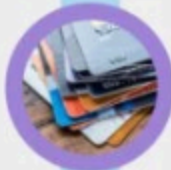


2018 • Healthcare fraud

600 people are implicated in the largest U.S. healthcare scam, which saw \$2 billion in false billings

2018 • Investment fraud

Elizabeth Holmes, founder of Theranos, is charged with massive fraud by the SEC for misrepresenting blood testing technology



2019 • Account fraud and takeovers

Over 100 million Capital One credit card applications and accounts are compromised

2019 • Bank and wire fraud

Huawei and its CFO are charged in connection with circumventing U.S. sanctions against Iran



2019 • Synthetic identities

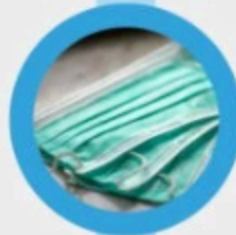
Police bust a fraud ring in New Jersey, which built up 7,000 credit profiles and created over \$200 million in losses

2020 • Invoice fraud

A Lithuanian hacker is jailed after tricking tech giants like Google into paying \$120 million in fake invoices

2020 • Disaster fraud

Fraudsters take advantage of the COVID-19 pandemic (phishing, fake masks and drugs, charity scams, false stimulus claims)



Trulioo

Don't be fooled. With Trulioo GlobalGateway, you can flag the fraudsters before they do any damage. Visit trulioo.com to learn more about the world's largest identity verification marketplace.

History of Phishing

<https://docs.google.com/document/d/1GHU013W1LNCo8eIdR6DQO69iVJKOFkwhHbeZ9vw9YaA/edit?tab=t.0>

- 1990s AOL internet phishing to get credentials and free internet
- Email to scam internet providers
- Love bug from Philippines infected 45 million windows pcs
- Why try to break a firewall when you can just socially engineer someone
- Technology will never fully stop phishing, old as time
- Phishing created by AOHell
 - https://publications.gc.ca/collections/collection_2021/cstc-csec/D96-69-2021-eng.pdf
- Paypal was getting impersonated
- Cryptocurrency makes scamming easier

History of Phishing Detection

- Defending against phishing attacks: taxonomy of methods, current issues and future directions
- https://books.google.ca/books?hl=en&lr=&id=ZjtaEQAAQBAJ&oi=fnd&pg=PA361&dq=history+of+phishing+detection&ots=MyLQd4usfA&sig=es_YBIfnhn--wmlbxNVd9RGaBIU#v=onepage&q=history%20of%20phishing%20detection&f=false

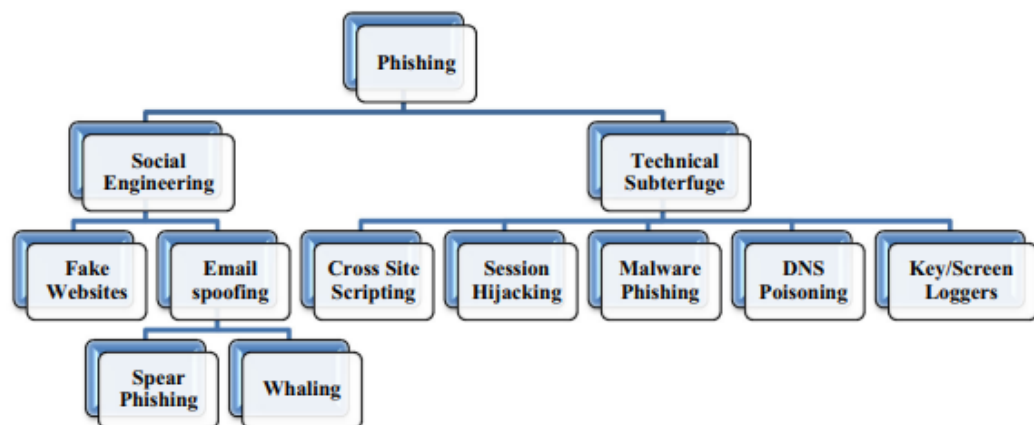


Fig. 8 Classification of phishing attacks based on how a phisher fraud to victims

- Defending against phishing attacks: taxonomy of methods, current issues and future directions

URL-based features	11	html-links	url_htmllink
	12	Number of dots in a link	url_nodots
	13	Non matching between target and text of urls	url_TarDiflink
	14	URL IP address	url_ip
	15	Image links	url_imagelink
	16	URL bag of word links	url_bagword
	17	URL has two domains	url_twodomain
	18	Non-standard port in the URL	url_nonstport
	19	URL containing hexadecimal characters or @ symbol	url_hexorat

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